

TẦNG 1

package "Presentation Layer" {

```
class MainMenu {
    + display_menu(): void
    + handle_login(): void
    + handle_register(): void
}

class AdminUI {
    + manage_movie(): void
    + manage_showtime(): void
    + manage_room(): void
    + view_statistics(): void
    + handle_logout(): void    // Để thoát trạng thái Admin, về MainMenu
}

class CustomerUI {
    - current_user: UserData
    + view_movie_list(): void
    + search_movie(): void
    + view_movie_detail(): void

    + select_seat(): void
    + select_showtime(): void

    + book_ticket(): void
    + cancel_ticket(): void

    + view_history(): void
    + view_showtimes(): void
    + print_ticket(): void
    + handle_logout(): void    // Để thoát trạng thái Customer, về MainMenu
}
}
```

TẦNG 2

package "Business Logic Layer" {

```
class AuthController {
    - user_table: UserHashTable
    + login(user: String, pass: String): String // Trả về "ADMIN", "CUSTOMER" hoặc
"FAILED" để UI chuyển hướng
    + register(user: String, pass: String): bool
    + logout(): void
    + check_role(username: String): String    // Kiểm tra quyền của tài khoản
}
```

```
}
```

```
class AdminController {  
    + calculate_revenue(): double  
    + count_movies(): int  
    + count_tickets(): int  
    + get_top_movie(): String  
    + generate_report(): void  
}
```

```
class MovieController {  
    - movie_list: MovieLinkedList  
  
    + add_movie(movie: MovieData): bool  
    + update_movie(movie_id: String, new_data: MovieData): bool  
    + delete_movie(movie_id: String): bool  
  
    + search_by_title(title: String): MovieNode  
    + sort_by_revenue(): void  
    + check_movie_booking_status(title: String): bool  
    + get_movie_list(): MovieLinkedList }  

```

```
class RoomController { Lớp điều phối phòng chiếu bị thiếu  
    - room_list: RoomLinkedList  
    + add_room(room: Room): bool  
    + delete_room(room_id: String): bool  
    + get_room_list(): RoomLinkedList  
    + find_room(room_id: String): RoomNode  
}
```

```
class ShowtimeController {  
    - showtime_list: ShowtimeLinkedList*  
  
    + add_showtime(st: Showtime): bool  
    + update_showtime(showtime_id: String, new_st: Showtime): bool  
    + delete_showtime(showtime_id: String): bool  
    + find_showtime(showtime_id: String): ShowtimeNode*  
  
    + get_movie_showtime(movie_id: String): String  
    + check_seat_status(showtime_id: String, r: int, c: int): bool  
    + reserve_seat(showtime_id: String, r: int, c: int): bool  
    + release_seat(showtime_id: String, r: int, c: int): void  
}
```

```
class BookingController {  
    - ticket_list: TicketLinkedList*  
    - showtime_controller: ShowtimeController // Chuyển sang dạng con trỏ đối tượng điều  
phối
```

```

+ process_booking(user: UserData, movie: MovieData, row: int, col: int): bool
+ cancel_booking(user_id: String, ticket_id: String): bool
+ find_ticket(ticket_id: String): String
+ get_booking_history(user_id: String): String
+ generate_ticket_info(ticket_id: String): String
}
}

```

TẦNG 3

package "Data Access & Structures" {

```

class FileIOHandler {
+ load_users(table: UserHashTable&): void
+ save_users(table: UserHashTable&): void
+ load_movies(list: MovieLinkedList&): void
+ save_movies(list: MovieLinkedList&): void
+ load_tickets(list: TicketLinkedList&): void
+ save_tickets(list: TicketLinkedList&): void
+ load_showtimes(list: ShowtimeLinkedList&): void
+ save_showtimes(list: ShowtimeLinkedList&): void
+ load_rooms(list: RoomLinkedList&): void    // Đọc ghi dữ liệu phòng chiếu
+ save_rooms(list: RoomLinkedList&): void    // Đọc ghi dữ liệu phòng chiếu
}

```

```

class UserNode {
- data: UserData
- next: UserNode*
+ get_data(): UserData
+ set_data(d: UserData): void
+ get_next(): UserNode*
+ set_next(nxt: UserNode*): void
}

```

```

class TicketNode {
- data: TicketData
- next: TicketNode*
+ get_data(): TicketData
+ get_next(): TicketNode*
}

```

```

class MovieNode {
- data: MovieData
- next: MovieNode
+ get_data(): MovieData
}

```

```
+ get_next(): MovieNode*  
}
```

```
class ShowtimeNode {  
    - data: Showtime  
    - next: ShowtimeNode*  
    + get_data(): Showtime  
    + get_next(): ShowtimeNode*  
}
```

```
class RoomNode {  
    - data: Room // Đồng bộ đặt tên thuộc tính là data cho giống các Node khác  
    - next: RoomNode*  
    + get_data(): Room  
    + get_next(): RoomNode*  
}
```

```
class UserHashTable {  
    - table: UserNode*[ ] // Mảng các con trỏ đầu danh sách liên kết vòng (Chaining)  
    - SIZE: int           // Kích thước mảng băm  
  
    + hash_function(key: String): int  
    + insert(username: String, value: UserData): void  
    + get(username: String): UserData  
    + contains(username: String): bool  
    + remove(username: String): bool  
    + display(): void  
}
```

```
class TicketLinkedList {  
    - head: TicketNode*  
    + add_ticket(ticket: TicketData): void  
    + traverse(): void  
    + remove_ticket(ticket_id: String): bool  
    + find_ticket(ticket_id: String): TicketNode*  
    + find_by_user(user_id: String): TicketLinkedList* }
```

```
class MovieLinkedList {  
    - head: MovieNode*  
    + add_movie(movie: MovieData): void  
    + remove_movie(movie_id: String): bool  
    + search_movie(title: String): MovieNode* + search_id(movie_id: String): MovieNode*  
    + sort_by_revenue_logic(): void  
    + traverse(): void  
}
```

```
class ShowtimeLinkedList {  
    - head: ShowtimeNode* + add_showtime(st: Showtime): void
```

```

    + find_showtime(showtime_id: String): ShowtimeNode* + find_by_movie(movie_id:
String): ShowtimeLinkedList*
    + traverse(): void
    + remove_showtime(showtime_id: String): bool //
}

class RoomLinkedList {
    - head: RoomNode* + add_room(room: Room): void
    + remove_room(room_id: String): bool
    + find_room(room_id: String): RoomNode* + traverse(): void
}
}

```

TẦNG 4:

package "Data Models" {

```

class UserData {
    - username: String
    - password: String
    - role: String
    - user_id: String
}

class TicketData {
    - ticket_id: String
    - user_id: String
    - movie_id: String
    - seat_id: String
    - status: String
    - showtime_id: String
    - room_id: String
    - price: double //để customer xem được giá vé ở mục lịch sử
}

class MovieData {
    - movie_id: String
    - title: String
    - genre: String
    - duration: int
    - description: String
    - revenue: double //doanh thu
    - base_price: double //giá sàn của phim
}

class SeatMatrix {
    - seats: int[ ][ ]
    - rows: int //Số hàng ghế của phòng
}

```

```

- cols: int      // Số cột ghế của phòng

+ check_status(r: int, c: int): int
+ reserve_seat(r: int, c: int): bool
+ book_seat(r: int, c: int): bool
+ release_seat(r: int, c: int): void
+ display_matrix(): void
}

class Showtime {
- showtime_id: String
- movie_id: String
- start_time: String
- room_id: String
- seat_matrix: SeatMatrix
+ get_available_seats(): int
+ display_showtime_info(): void
}

class Room {
- room_id: String
- room_name: String
- capacity: int
}
}

```